Enzyme	Optimum temperature (^{0}C)
Plant urease	60
Human enzymes	37

But enzymes like venom phosphokinase, muscle adenylate kinase are active even at 100^{0}C.

5. In general, at high temperatures the enzymes undergo denaturation and this leads to the rapid loss of catalytic activity.

6. Temperature co-efficient or Q_{10} is defined as increase in velocity of the enzymatic reaction when the temperature is increased by 10^0 C. For most of the enzymes the value of Q_{10} is "2" in the temperature range 0^0 C to 40^0 C.

4.4.3 Concentration of substrate:

Formation of an enzyme-substrate complex (ES complex) is the first step in enzyme catalysis. Increase in the substrate concentration gradually increases the velocity of enzymatic reaction up to a particular value.

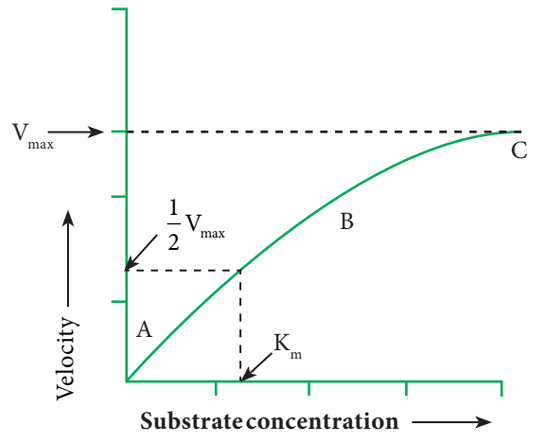


Figure 4.5 Effect of substrate concentration

A hyperbolic curve is obtained when velocity is plotted against the substrate concentration. This graph shows three distinct phases.

i) In the first phase (A), the velocity of the reaction is directly proportional to the substrate concentration.

ii) In the second phase (B), the substrate concentration is not directly proportional to the enzyme activity.

iii) In the third phase (C), the velocity remains constant and does not change with increase in the substrate concentration. (plateau)

4.4.4 Concentration of enzyme

At a constant substrate concentration, the velocity of enzyme catalyzed reaction increases proportionately with the increase in the concentration of the enzyme. This property is utilized in determining the level of serum enzymes for the diagnosis of diseases. On plotting the velocity of the enzymatic reaction with the enzyme concentration, a straight line is obtained.

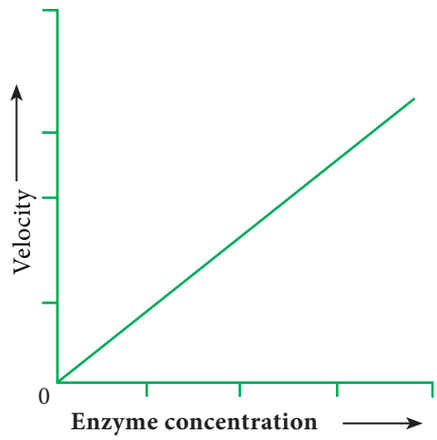


Figure 4.6 Effect of enzyme concentration

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